

Chapter 1

Density Estimation

1.1 Density Estimation

$f_x \rightarrow$ parameter (infinite dimensional parameter) of interest

1. $f_x \geq 0$
2. $\int f(x) dx = 1$
3. $\int_a^b f(x) dx = P(a < X < b)$

$X_1, \dots, X_n \sim f_x$ have distribution same as X .

$\hat{f} \rightarrow$ histogram. Area under the curve as integral, then indeed $\int \hat{f}(x) dx = 1$.

$\hat{f} \geq 0$

For a, b endpoints of a bin,

$$\int_a^b \hat{f}(x) dx = \frac{\#\{X_i \in (a, b)\}}{n} = \hat{F}_n(b) - \hat{F}_n(a) = \text{empirical probability.}$$

Histogram satisfies (1)–(3).

Problems

1. It is not smooth.
2. Location and width of bins is subjective. (not a function of data only).

Minimize MSE to find optimal bin width (bandwidth).

Find MSE. $x_0, x_1, x_2, \dots, x_n \rightarrow$ endpoints of bins.

$$\begin{aligned} \hat{f}(t) &= \begin{cases} \frac{\hat{F}_n(x + \delta) - \hat{F}_n(x)}{\delta} & x_i < t < x_{i+1} \\ 0 & \text{otherwise} \end{cases} \\ x_i &= x, \quad x_{i+1} = x + \delta \\ \hat{f}(t) &= \frac{\hat{F}_n(x + \delta) - \hat{F}_n(x)}{\delta} = \frac{\#\{X_i \in (x, x + \delta)\}}{n\delta} \end{aligned}$$

$$n\hat{F}_n(x) \sim \text{Bin}(n, F(x)) \quad \sim \frac{1}{n\delta} \text{Bin}(n, F(x + \delta) - F(x))$$



Figure 1.1: Step Function Diagram

$$F(x + \delta) - F(x) = p$$

$$E\hat{f}(t) = \frac{1}{n\delta} \cdot n[F(x + \delta) - F(x)] = \frac{1}{\delta}[F(x + \delta) - F(x)] = \frac{p}{\delta}$$

$$\text{Var } \hat{f}(t) = \frac{1}{n^2\delta^2} np(1-p) = \frac{p(1-p)}{n\delta^2}$$

$$\text{MSE} = \text{Bias}^2 + \text{Var} = \left(\frac{p}{\delta} - f(t)\right)^2 + \frac{p(1-p)}{n\delta^2}$$

Minimize thus over δ to get bin width.

$$\left[\begin{array}{l} \text{Best choice is } \delta \propto n^{-1/5}. \text{ Larger sample size should have} \\ \text{denser (more bins) histograms.} \\ \text{MSE} = O(n^{-4/5}) \text{ not } 1/n. \quad \text{MSE}(x) = \frac{\sigma^2}{n} \end{array} \right.$$

Location – Shifted histogram. (fix δ).

Actual evaluation does not need to be done at every ($\mathbb{R}$) points, but only at discrete set of points.

$$\hat{f}(t) = \#\{X_i \in (t - \delta/2, t + \delta/2)\} / \delta$$

Smoothness: Kernel density.

$K(z)$ is a standard kernel function if

- K is Symmetric around zero $k(z) = K(-z)$
- $K(z) \geq 0 \forall z \quad \int_{-\infty}^{\infty} K(z) dz = 1$
- The mass function is non decreasing on both sides of zero.
- $\int z k(z) dz = 0 \quad \int z^2 k(z) dz = 1$

Bandwidth re-scales kernel

$$k_h(z) = \frac{1}{h} K\left(\frac{z}{h}\right)$$

e.g. Rectangular $K(z) = \frac{1}{2}\mathbf{1}(-1 < z < 1)$



Figure 1.2: smooth bump-shaped density estimate over data points $x_1, x_2, \dots$

Gaussian $K(z) = \frac{1}{\sqrt{2\pi}} e^{-z^2/2}$

Triangular $K(z) = (1 - |z|) \mathbf{1}(-1 < z < 1)$

Epanechnikov $K(z) = \frac{3}{4}(1 - z^2) \mathbf{1}(-1 < z < 1)$

Kernel density estimators

$$\hat{f}_h(t) = \frac{1}{n} \sum_{i=1}^n k_h(X_i - t)$$

Shifted histogram is a kernel density estimate with rectangular kernel. The height of the shifted histogram at a point t is proportional to the # of data points at a distance less than h to t .

Convolution of two densities f_x, g_y

$$f_S(z) = f_x * g_y(z) = \int_{-\infty}^{\infty} f_x(t) g_y(z - t) dt = \int_{-\infty}^{\infty} f_x(z - t) g_y(t) dt$$

f_S is the density of a r.v. $S = X + Y$.

Convolution of 2 pmfs p_x and p_y

$$p_S(x) = p_x * p_y(x) = \sum_{t \in \mathbb{R}} p_x(t) p_y(x - t)$$

p_S is the pmf of $S = X + Y$

X independent of Y

f_x is density of X and p_y is the pmf of Y .

$S = X + Y$ is continuous.

$$P(S \leq a) = \sum_{k \in \mathbb{R}} P(Y = k \text{ and } X \leq a - k) = \sum_{k \in \mathbb{R}} p_y(k) F_x(a - k)$$

$$f_S(a) = \sum_{t \in \mathbb{R}} p_y(t) f_x(a - t)$$

Back to kernels.

$$\hat{f}(t) = \frac{1}{n} \sum_{i=1}^n k_h(t - X_i) = \sum_{i=1}^n p_y(X_i) k_h(t - X_i)$$

p_y is the pmf that puts mass $\frac{1}{n}$ at each $X_i, i = 1, \dots, n$.

$$p_y(t) = \begin{cases} \frac{1}{n} & \text{if } t \in \{x_1, \dots, x_n\} \\ 0 & \text{o.w.} \end{cases}$$

So if k_h is density of a cont r.v. then $\hat{f}(t)$ is also a density.

Bandwidth selection

Minimize MSE.

$$\hat{f}_h(x) = \frac{1}{n} \sum_{i=1}^n k_h(x - X_i) \quad k_h(t) = \frac{1}{h} k\left(\frac{t}{h}\right) \quad \int_{\mathbb{R}} k(u) du = 1$$

$$\begin{aligned} E\left(\hat{f}_h(x)\right) &= \frac{1}{n} \sum_{i=1}^n E(k_h(x - X_i)) \quad X_i \sim f \\ &= E(k_h(x - X_1)) \end{aligned}$$

$$\text{Var}(\hat{f}_h(x)) = \frac{1}{n^2} \sum_{i=1}^n \text{Var}(k_h(x - X_i)) = \frac{1}{n} \text{Var}(k_h(x - X_1))$$

$$E(k_h(x - X_1)) = \int k_h(x - t) f(t) dt \quad k_h(x) \simeq 0 \text{ for } |x| > h$$

$$\text{Taylor expansion} \quad f(t) = f(x) + (t - x) f'(x) + \frac{(t - x)^2}{2} f''(x) + o(h^2)$$

$$\begin{aligned} E(k_h(x - X_1)) &= \int k_h(x - t) f(x) dt + \int k_h(x - t) (t - x) f'(x) dt + \int k_h(x - t) \frac{(t - x)^2}{2} f''(x) dt \\ &= f(x) \cdot 1 + f'(x) \int (t - x) k_h(t - x) dt + \text{smaller terms} \\ &\quad + \frac{f''(x)}{2} \int (t - x)^2 k_h(t - x) dt + \text{smaller terms} \end{aligned}$$

$$E(\hat{f}|x) = f(x) + 0 + \frac{h^2}{2} f''(x) + o(h^2)$$

$$\text{Bias} = \frac{h^2}{2} f''(x) + o(h^2).$$

$$\begin{aligned} E(k_h^2(x - X_1)) &= \int k_h^2(x - t) f(t) dt \\ &= \int \left[\frac{1}{h} k\left(\frac{x - t}{h}\right) \right]^2 f(t) dt \\ &= \frac{1}{h^2} \int k^2\left(\frac{x - t}{h}\right) f(t) dt \quad \frac{x - t}{h} = u \\ &= \frac{1}{h^2} \cdot h \int k^2(u) f(x + hu) du \\ &= \frac{1}{h} \left[\int k^2(u) f(x) du + \text{smaller terms} \right] \end{aligned}$$

$$\begin{aligned} \text{Var}(\hat{f}_h(x)) &= \frac{1}{n} \left[\frac{1}{h} f(x) \cdot \int k^2(u) du + \text{smaller terms} - (E\hat{f})^2 \right] \\ &= \frac{1}{nh} \int f(x) \cdot k^2(u) du + \text{smaller terms.} \end{aligned}$$

$$\text{Bias} = \frac{h^2}{2} f''(x)$$

$$\text{MSE} = h^4 C_1(x) + \frac{1}{nh} C_2(x) + \text{smaller terms.}$$

$$h^4 = O\left(\frac{1}{nh}\right) \quad \text{To minimize MSE}$$

$$h \propto n^{-1/5}$$

$$h^4 = n^{-4/5}$$

$$\text{MSE} = O(n^{-4/5})$$

$$\frac{1}{nh} = n^{-1}h^{-1} = n^{-4/5}$$

$h < O(n^{-1/5})$ then $\frac{1}{nh}$ is of higher order than $n^{-4/5}$

$h > O(n^{-1/5})$ then h^4 is of higher order than $n^{-4/5}$

Normal reference rule.

If f is normal then

$$h = 1.06 \sigma n^{-1/5}$$

= One can estimate σ from the data.

If the histogram is symmetric unimodal then this choice of h can be used.

Cross-validation to determine h

$$E\left(\hat{f}_h(x) - f(x)\right)^2$$

minimise $\int \left(\hat{f}_h(x) - f(x)\right)^2 dx$.

$$\begin{aligned} &= \int \hat{f}_h^2(x) dx - 2 \int \hat{f}_h(x) f(x) dx + \int f^2(x) dx \\ &= F_1 + F_2 + F_3 \end{aligned}$$

For a particular h , we can compute $\hat{f}_h(x)$ for every x and hence approximate the first term.

The third term does not contain h , so it does not affect the choice of h .

Second term involves h (so we do need it) and the unknown f .

$$-2 \int \hat{f}_h(x) f(x) dx = -2 E \hat{f}_h(X) \quad \text{where } X \sim f.$$

You have $X_1, \dots, X_n$ i.i.d. f .

Estimate $E \hat{f}_h(x)$ by $\frac{1}{n} \sum_{i=1}^n \hat{f}_h(X_i)$.

Then minimise the approximation to F_1 (numerically) and (the estimate of F_2) (numerically) to get h .

$$\boxed{\frac{1}{n} \sum_{i=1}^n \hat{f}_h(X_i)} \rightarrow \text{is biased for } E\hat{f}_h(x)$$

$$\frac{1}{n} \sum_{i=1}^n \frac{1}{n} \sum_{j=1}^n k_h(X_i - X_j)$$

$$E(k_h(X_i - X_j)) = E\hat{f}_h(X_i) \quad \text{for } i \neq j$$

$$i = j \quad E(k_h(X_i - X_j)) = E(k_h(0)) = k_h(0).$$

$$E \left[\frac{1}{n} \sum_{j=1}^n k_h(X_i - X_j) \right] = \frac{1}{n} \left[(n-1) E\hat{f}_h(x) + k_h(0) \right] = E\hat{f}_h(X_i) + \frac{1}{n} \left(k_h(0) - E\hat{f}_h(x) \right)$$

$$E \left[\frac{1}{n} \sum_{j=1}^n k_h(X_i - X_j) \right] = \frac{1}{n} \left[(n-1) E\hat{f}_h(x_i) + k_h(0) \right] = E\hat{f}_h(X_i) + \underbrace{\frac{1}{n} \left(k_h(0) - E\hat{f}_h(x) \right)}_{\text{bias.}}$$

Correct this by removing $k_h(X_i - X_i)$

$$\hat{f}_{-i}(X_i) = \frac{1}{n-1} \sum_{j \neq i} k_h(X_i - X_j)$$

$$(-2) \frac{1}{n} \sum_{i=1}^n \hat{f}_{-i}(X_i) \rightarrow \text{estimates } F_2. \quad \text{Leave one out cross validation}$$